

Exponentials & Logarithms

ANSWER KEY



Exponent rules

- 1 Simplify $x^5 \cdot x^3$.

$$x^{5+3} = x^8.$$

- 2 Simplify $(2x^3)^4$.

$$2^4 x^{12} = 16x^{12}.$$

- 3 Simplify $\frac{5^6}{5^2}$.

$$5^{6-2} = 5^4 = 625.$$

Solving exponential equations

- 4 Solve $3^x = 81$.

$$81 = 3^4, \text{ so } x = 4.$$

- 5 Solve $2^{x+3} = 64$.

$$64 = 2^6, \text{ so } x + 3 = 6, \text{ giving } x = 3.$$

- 6 Solve $5^{2x} = 125$.

$$125 = 5^3, \text{ so } 2x = 3, \text{ giving } x = \frac{3}{2}.$$

Logarithm definition and evaluation

- 7 Evaluate $\log_2 64$.

$$2^6 = 64, \text{ so } \log_2 64 = 6.$$

- 8 Evaluate $\log_5 1$.

$$5^0 = 1, \text{ so } \log_5 1 = 0.$$

- 9 Rewrite $\log_3 81 = 4$ in exponential form, and verify.

$$3^4 = 81, \text{ which checks out.}$$

Logarithm laws

- 10 Simplify $\log 8 + \log 5$ as a single logarithm.

$$\log(8 \times 5) = \log 40.$$

- 11 Simplify $\log_2 48 - \log_2 3$.

$$\log_2 \left(\frac{48}{3} \right) = \log_2 16 = 4.$$

- 12 Write $3\log_2 x + \log_2 y$ as a single logarithm.

$$\log_2 (x^3 y).$$

Solving logarithmic equations

- 13 Solve $\log_4 (x + 5) = 2$.

$$4^2 = 16 = x + 5, \text{ so } x = 11.$$

- 14 Solve $\log_2 (x) + \log_2 (x - 2) = 3$.

$\log_2 [x(x - 2)] = 3$, so $x(x - 2) = 8$, giving $x^2 - 2x - 8 = 0$, i.e. $(x - 4)(x + 2) = 0$. So $x = 4$ or $x = -2$; but $x = -2$ makes the original logarithms undefined, so the only valid solution is $x = 4$.

Exponential growth and decay models

- 15 A population of bacteria starts at 300 and doubles every 4 hours. Write an exponential model and find the population after 12 hours.

$$P(t) = 300 \cdot 2^{t/4}. P(12) = 300 \cdot 2^3 = 300 \times 8 = 2400.$$

- 16 A car worth 24,000 dollars depreciates at 12 percent per year. Find its value after 3 years, to the nearest dollar.

$$V = 24000(0.88)^3 = 24000 \times 0.681472 \approx 16,355 \text{ dollars}.$$

Graphs of exponential functions

- 17 The graph of $y = 2^x$ is shown below. Use the graph to estimate the value of x for which $y = 8$, then confirm your answer algebraically.

From the graph, $x \approx 3$. Algebraically: $2^x = 8 = 2^3$, so $x = 3$.

Natural logarithms and e

- 18 Evaluate $\ln(e^5)$.

Since $\ln$ and e^x are inverse functions, $\ln(e^5) = 5$.

- 19 Solve $e^x = 20$ for x , to the nearest hundredth, using natural logarithms.

$$x = \ln(20) \approx 2.996 \approx 3.00.$$

A well-designed word problem: comparing two investment accounts

- 20 This question has three parts. An investment of 5000 dollars is placed in an account that grows according to $A(t) = 5000e^{0.04t}$, where t is measured in years. (a) Find the value of the investment after 10 years, to the nearest dollar. (b) Find how many years it takes for the investment to double, to the nearest tenth of a year, using natural logarithms. (c) A second account offers simple (non-compounded) growth of 5 percent per year on the same initial 5000 dollars. Determine which account has the greater value after 10 years, and by how much, to the nearest dollar.

(a) $A(10) = 5000e^{0.4} \approx 5000(1.4918) \approx 7459$ dollars. (b) Set $10000 = 5000e^{0.04t}$, so $2 = e^{0.04t}$, giving $\ln 2 = 0.04t$, so $t = \frac{\ln 2}{0.04} \approx \frac{0.6931}{0.04} \approx 17.3$ years. (c) Simple growth value: $5000 + 5000(0.05)(10) = 5000 + 2500 = 7500$ dollars. Since $7500 > 7459$, the simple-growth account is greater after 10 years, by about $7500 - 7459 = 41$ dollars.

Multiple choice — easy

- 21 Solve $6^x = 216$.

a. 3 ✓

b. 216

c. 4

d. 2

$$216 = 6^3, \text{ so } x = 3.$$

22 Solve $2^x = 32$.

a. 4

b. 6

c. 5 ✓

d. 32

$$32 = 2^5, \text{ so } x = 5.$$

23 Solve $4^x = 16$.

a. 16

b. 2 ✓

c. 1

d. 3

$$16 = 4^2, \text{ so } x = 2.$$

24 Solve $10^x = 1000$.

a. 3 ✓

b. 2

c. 4

d. 1000

$$1000 = 10^3, \text{ so } x = 3.$$

25 Solve $5^x = 125$.

a. 3 ✓

b. 125

c. 4

d. 2

$$125 = 5^3, \text{ so } x = 3.$$

26 Solve $7^x = 2401$.

- a. 4 ✓
- b. 3
- c. 2401
- d. 5

$2401 = 7^4$, so $x = 4$.

27 Solve $3^x = 9$.

- a. 1
- b. 9
- c. 2 ✓
- d. 3

$9 = 3^2$, so $x = 2$.

28 Solve $5^x = 25$.

- a. 25
- b. 2 ✓
- c. 3
- d. 1

$25 = 5^2$, so $x = 2$.

29 Solve $2^x = 16$.

a. 16

b. 5

c. 4 ✓

d. 3

$16 = 2^4$, so $x = 4$.

30 Solve $4^x = 64$.

a. 3 ✓

b. 4

c. 64

d. 2

$64 = 4^3$, so $x = 3$.

31 Solve $3^x = 27$.

a. 2

b. 27

c. 4

d. 3 ✓

$27 = 3^3$, so $x = 3$.

32 Solve $2^x = 4$.

a. 3

b. 4

c. 1

d. 2 ✓

$4 = 2^2$, so $x = 2$.

33 Solve $9^x = 59049$.

a. 4

b. 59049

c. 6

d. 5 ✓

$59049 = 9^5$, so $x = 5$.

34 Solve $9^x = 729$.

a. 3 ✓

b. 4

c. 2

d. 729

$729 = 9^3$, so $x = 3$.

35 Solve $5^x = 625$.

a. 3

b. 4 ✓

c. 5

d. 625

$625 = 5^4$, so $x = 4$.

Multiple choice — medium

36 Solve $\log_3(x + 3) = 4$.

- a. 80
- b. 76
- c. 81
- d. 78 ✓

$$3^4 = 81 = x + 3, \text{ so } x = 78.$$

37 Solve $\log_2(x + 7) = 2$.

- a. -1
- b. -3 ✓
- c. -5
- d. 4

$$2^2 = 4 = x + 7, \text{ so } x = -3.$$

38 Solve $\log_5(x + 15) = 2$.

- a. 25
- b. 12
- c. 8
- d. 10 ✓

$$5^2 = 25 = x + 15, \text{ so } x = 10.$$

39 Solve $\log_4(x - 12) = 1$.

- a. 18
- b. 14
- c. 4
- d. 16 ✓

$$4^1 = 4 = x - 12, \text{ so } x = 16.$$

40 Solve $\log_4(x + 12) = 1$.

a. -8 ✓

b. -6

c. -10

d. 4

$$4^1 = 4 = x + 12, \text{ so } x = -8.$$

41 Solve $\log_4(x + 5) = 1$.

a. 1

b. -3

c. 4

d. -1 ✓

$$4^1 = 4 = x + 5, \text{ so } x = -1.$$

42 Solve $\log_2(x - 10) = 1$.

a. 10

b. 12 ✓

c. 14

d. 2

$$2^1 = 2 = x - 10, \text{ so } x = 12.$$

43 Solve $\log_5(x + 9) = 2$.

a. 16 ✓

b. 14

c. 25

d. 18

$$5^2 = 25 = x + 9, \text{ so } x = 16.$$

44 Solve $\log_2(x + 4) = 2$.

a. 4

b. 2

c. -2

d. 0 ✓

$$2^2 = 4 = x + 4, \text{ so } x = 0.$$

45 Solve $\log_4(x - 1) = 3$.

a. 63

b. 64

c. 65 ✓

d. 67

$$4^3 = 64 = x - 1, \text{ so } x = 65.$$

46 Solve $\log_3(x - 6) = 1$.

a. 11

b. 3

c. 9 ✓

d. 7

$$3^1 = 3 = x - 6, \text{ so } x = 9.$$

47 Solve $\log_4(x + 15) = 3$.

a. 51

b. 47

c. 49 ✓

d. 64

$$4^3 = 64 = x + 15, \text{ so } x = 49.$$

48 Solve $\log_5(x - 13) = 2$.

a. 25

b. 40

c. 36

d. 38 ✓

$$5^2 = 25 = x - 13, \text{ so } x = 38.$$

49 Solve $\log_3(x + 5) = 4$.

a. 76 ✓

b. 78

c. 81

d. 74

$$3^4 = 81 = x + 5, \text{ so } x = 76.$$

50 Solve $\log_4(x + 13) = 4$.

- a. 256
- b. 241
- c. 245
- d. 243 ✓

$$4^4 = 256 = x + 13, \text{ so } x = 243.$$

Multiple choice — hard

51 A population of bacteria starts at 488 and doubles every 6 hours. Find the population after 12 hours.

- a. 2440
- b. 1952 ✓
- c. 976
- d. 2928

$$P(t) = 488 \cdot 2^{t/6}. P(12) = 488 \cdot 2^2 = 488 \times 4 = 1952.$$

52 A population of bacteria starts at 497 and doubles every 3 hours. Find the population after 12 hours.

- a. 1988
- b. 3976
- c. 7952 ✓
- d. 8449

$$P(t) = 497 \cdot 2^{t/3}. P(12) = 497 \cdot 2^4 = 497 \times 16 = 7952.$$

53 A population of bacteria starts at 160 and doubles every 4 hours. Find the population after 16 hours.

a. 2560 ✓

b. 640

c. 2720

d. 1280

$$P(t) = 160 \cdot 2^{t/4}. P(16) = 160 \cdot 2^4 = 160 \times 16 = 2560.$$

54 A population of bacteria starts at 351 and doubles every 2 hours. Find the population after 8 hours.

a. 5616 ✓

b. 2808

c. 1404

d. 5967

$$P(t) = 351 \cdot 2^{t/2}. P(8) = 351 \cdot 2^4 = 351 \times 16 = 5616.$$

55 A population of bacteria starts at 142 and doubles every 2 hours. Find the population after 4 hours.

a. 710

b. 568 ✓

c. 284

d. 852

$$P(t) = 142 \cdot 2^{t/2}. P(4) = 142 \cdot 2^2 = 142 \times 4 = 568.$$

56 Solve $e^x = 671$ for x , to the nearest hundredth, using natural logarithms.

a. 6.51 ✓

b. 7.01

c. 7.51

d. 6.01

$$x = \ln(671) \approx 6.51.$$

57 Solve $e^x = 898$ for x , to the nearest hundredth, using natural logarithms.

a. 6.80 ✓

b. 6.30

c. 7.80

d. 7.30

$$x = \ln(898) \approx 6.80.$$

58 Solve $e^x = 144$ for x , to the nearest hundredth, using natural logarithms.

a. 5.97

b. 4.97 ✓

c. 5.47

d. 4.47

$$x = \ln(144) \approx 4.97.$$

59 Solve $e^x = 468$ for x , to the nearest hundredth, using natural logarithms.

a. 5.65

b. 6.65

c. 6.15 ✓

d. 7.15

$$x = \ln(468) \approx 6.15.$$

60 Solve $e^x = 971$ for x , to the nearest hundredth, using natural logarithms.

a. 7.88

b. 7.38

c. 6.38

d. 6.88 ✓

$$x = \ln(971) \approx 6.88.$$

61 Simplify $\log_5 9375 - \log_5 3$.

a. 4

b. 6

c. 5 ✓

d. 9372

$$\log_5 \left(\frac{9375}{3} \right) = \log_5 3125 = 5.$$

62 Simplify $\log_3 567 - \log_3 7$.

a. 4 ✓

b. 3

c. 560

d. 5

$$\log_3 \left(\frac{567}{7} \right) = \log_3 81 = 4.$$

63 Simplify $\log_5 150 - \log_5 6$.

a. 3

b. 144

c. 2 ✓

d. 1

$$\log_5 \left(\frac{150}{6} \right) = \log_5 25 = 2.$$

64 Simplify $\log_2 112 - \log_2 7$.

a. 5

b. 105

c. 3

d. 4 ✓

$$\log_2 \left(\frac{112}{7} \right) = \log_2 16 = 4.$$

65 Simplify $\log_3 405 - \log_3 5$.

a. 4 ✓

b. 400

c. 3

d. 5

$$\log_3 \left(\frac{405}{5} \right) = \log_3 81 = 4.$$